

Partial Derivatives

Function of multiple variable

Function of two variable: A function of two variable, x and y , is a rule that assigns a unique real number $f(x, y)$ to each point (x, y) in some set D in the xy -plane.

Function of three variable: A function of two variable, x, y and z , is a rule that assigns a unique real number $f(x, y, z)$ to each point (x, y, z) in some set D in three-dimensional plane.

Example-1:

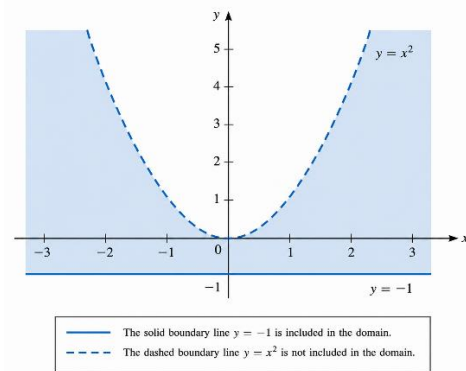
Let $f(x, y) = \sqrt{y+1} + \ln(x^2 - y)$. Find $f(e, 0)$ and sketch the natural domain of the function.

Solution: By substitution,

$$f(e, 0) = \sqrt{1} + \ln(e^2) = 1 + 2 = 3$$

To find the natural domain of f , $\sqrt{y+1}$ we note that is defined only when $y \geq -1$, while $\ln(x^2 - y)$ is defined only when $0 < (x^2 - y)$ or $y < x^2$. Thus, the natural domain of f consists of all points in the xy -plane for which $-1 \leq y < x^2$. To sketch the natural domain,

we first sketch the parabola $y = x^2$ as a “dashed” curve and the line $y = -1$ as a solid curve. The natural domain of f is then the region lying above or on the line $y = -1$ and below the parabola $y = x^2$.



Example-2: Let $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$. Find $f\left(0, \frac{1}{2}, -\frac{1}{2}\right)$ and the natural domain of f .

Solution: By substitution,

$$f\left(0, \frac{1}{2}, -\frac{1}{2}\right) = \sqrt{\frac{1}{2}}$$

Because of square root sign we must have, $0 \leq 1 - x^2 - y^2 - z^2$ in order to have real value of $f(x, y, z)$. Rewriting the form

$$x^2 + y^2 + z^2 \leq 1$$

Therefor we see that the natural domain of f consists of all points on or within the sphere

$$x^2 + y^2 + z^2 = 1$$

Example-3: In each part describe the graph of the function in xyz-coordinate system.

a) $f(x, y) = 1 - x - \frac{1}{2}y$

b) $f(x, y) = \sqrt{1 - x^2 - y^2}$.

Solution: (a) By the definition, the graph of given function is the graph of the equation

$$z = 1 - x - \frac{1}{2}y$$

Which is a plane. A triangular portion of this plane can be sketched by plotting the intersections with the coordinate axes and joining them with the line segments. The intersections with the coordinates are $(1,0,0)$, $(0,2,0)$, $(0,0,1)$ respectively.

(b) By the definition, the graph of given function is the graph of the equation

$$z = \sqrt{1 - x^2 - y^2} \quad (1)$$

After squaring both sides we get

$$x^2 + y^2 + z^2 = 1$$

Which represents a sphere of radius 1, centered at origin. Since (2) imposes the condition $z \geq 0$, the graph is just the upper hemisphere.

Limit and Continuity:

Limit of functions of two variables: Let f be a function of two variables, and assume that f is defined at all points of some open disk centered at (x_0, y_0) , except possibly at (x_0, y_0) . We will write

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$$

if given any number $\varepsilon > 0$, we can find a number $\delta > 0$ such that $f(x, y)$ satisfies

$$|f(x, y) - L| < \varepsilon$$

whenever the distance between (x, y) and (x_0, y_0) satisfies

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

Limit of functions of three variables: Let f be a function of three variables, and assume that f is defined at all points of some open disk centered at (x_0, y_0, z_0) , except possibly at (x_0, y_0, z_0) . We will write

$$\lim_{(x,y,z) \rightarrow (x_0,y_0,z_0)} f(x, y, z) = L$$

if given any number $\varepsilon > 0$, we can find a number $\delta > 0$ such that $f(x, y, z)$ satisfies

$$|f(x, y, z) - L| < \varepsilon$$

whenever the distance between (x, y, z) and (x_0, y_0, z_0) satisfies

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} < \delta$$

Example-4: Evaluate $\lim_{(x,y) \rightarrow (-1,2)} \frac{xy}{x^2+y^2}$

$$\begin{aligned} \text{Solution: } & \lim_{(x,y) \rightarrow (-1,2)} \frac{xy}{x^2+y^2} \\ &= \frac{(-1)2}{1+4} \\ &= \frac{-2}{5} \end{aligned}$$

Example-5: Evaluate $\lim_{(x,y) \rightarrow (1,4)} (5x^3y^2 - 9)$

$$\text{Solution: } \lim_{(x,y) \rightarrow (1,4)} (5x^3y^2 - 9)$$

$$= \lim_{(x,y) \rightarrow (1,4)} (5x^3y^2) - \lim_{(x,y) \rightarrow (1,4)} 9$$

$$= 5 \left[\lim_{(x,y) \rightarrow (1,4)} x^3 \lim_{(x,y) \rightarrow (1,4)} y^2 \right] - 9$$

$$= 5 \times 1 \times 16 - 9 = 71$$

Theorem:

(a) If $f(x, y) \rightarrow L$ as $(x, y) \rightarrow (x_0, y_0)$, then $f(x, y) \rightarrow L$ as $(x, y) \rightarrow (x_0, y_0)$, along any smooth curve.

(b) If the limit of $f(x, y)$ fails to exist as $(x, y) \rightarrow (x_0, y_0)$ along some smooth curve, or if $f(x, y)$ has different limits as $(x, y) \rightarrow (x_0, y_0)$ along two different smooth curves, then the limit of $f(x, y)$ does not exist as $(x, y) \rightarrow (x_0, y_0)$.

Example-6: $\lim_{(x,y) \rightarrow (0,0)} -\frac{xy}{x^2+y^2}$. Does the limit exist?

Solution: $\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } x=0}} -\frac{xy}{x^2+y^2} = 0$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=0}} -\frac{xy}{x^2+y^2} = 0$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=x}} -\frac{xy}{x^2+y^2} = -\frac{1}{2}$$

We found two different smooth curves along which this limit had different values. Therefore the limit doesn't exist.

Example-7: Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x^2+y^2}$

Solution:

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } x=0}} \frac{x-y}{x^2+y^2} = \lim_{y \rightarrow 0} \frac{-y}{y^2} = -\infty$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=0}} \frac{x-y}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{x}{x^2} = \infty$$

Therefore $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x^2+y^2}$ doesn't exist.

Continuity of functions of two variable: A function $f(x, y)$ is said to be **continuous at** (x_0, y_0) if $f(x_0, y_0)$ is defined and if

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(x_0, y_0)$$

In addition, if f is continuous at every point in an open set D , then we say that f is continuous on D , and if f is continuous at every point in the xy -plane, then we say that f is continuous everywhere.

Example-8:

$$f(x, y) = \begin{cases} \frac{\sin(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$$

Show that f is continuous at $(0, 0)$

Solution: To show that

$$f(x, y) = \begin{cases} \frac{\sin(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$$

Is continuous at $(0, 0)$ we must to proof that $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0) = 1$

$$\lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } x=0}} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \lim_{y \rightarrow 0} \frac{\sin y^2}{y^2} = 1$$

$$\lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } y=0}} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = 1$$

$$\lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } y=x}} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{\sin(2x^2)}{2x^2} = 1$$

$$\text{Therefor } \lim_{(x, y) \rightarrow (0, 0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = 1$$

$$\text{Therefor } \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0) = 1$$

Hence the function continuous at $(0, 0)$.

Example-9:

$$f(x, y) = \begin{cases} \frac{x + y}{2x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Show that f is discontinuous at $(0, 0)$

Solution: To show that

$$f(x, y) = \begin{cases} \frac{x+y}{2x^2+y^2} & \text{if } (x, y) \neq (0,0) \\ 0 & \text{if } (x, y) = (0,0) \end{cases}$$

Is continuous at (0,0) we must to proof that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = f(0,0) = 1$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } x=0}} \frac{x+y}{2x^2+y^2} = \lim_{y \rightarrow 0} \frac{y}{y^2} = \infty$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=0}} \frac{x+y}{2x^2+y^2} = \lim_{y \rightarrow 0} \frac{x}{2x^2} = \infty$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=-x}} \frac{x-x}{2x^2+x^2} = 0$$

Therefor $\lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{2x^2+y^2}$ doesn't exist

Therefor f is discontinuous at (0,0).

Partial Differentiation:

Example-10: $u = x^3 - 3x^2y + 3xy^2 - y^3$, find $\frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial y^2}, \frac{\partial^2 u}{\partial x \partial y}, \frac{\partial^2 u}{\partial y \partial x}$

Solution: Given $u = x^3 + 3x^2y + 3xy^2 + y^3$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x^3 + 3x^2y + 3xy^2 + y^3)$$

$$= 3x^2 + 6xy + 3y^2$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} (3x^2 + 6xy + 3y^2)$$

$$= 6x + 6y$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (x^3 + 3x^2y + 3xy^2 + y^3)$$

$$= 3x^2 + 6xy + 3y^2$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right)$$

$$= \frac{\partial}{\partial y} (3x^2 + 6xy + 3y^2)$$

$$= 6x + 6y$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} (3x^2 + 6xy + 3y^2)$$

$$= 6x + 6y$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right)$$

$$= \frac{\partial}{\partial y} (3x^2 + 6xy + 3y^2)$$

$$= 6x + 6y$$

Chain Rule:

Chain Rule for Derivative: If $x = x(t)$ and $y = y(t)$ are differentiable at t , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(t), y(t))$, then $z = (x(t), y(t))$, is differentiable at t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y)

Example-11: Suppose that, $z = x^2y$, $x = t^2$, $y = t^3$. Find $\frac{dz}{dt}$ by using chain rule.

Solution:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\frac{dz}{dt} = 2xy \frac{dx}{dt} + x^2 \frac{dy}{dt}$$

$$\frac{dz}{dt} = 2xy \frac{d}{dt}(t^2) + x^2 \frac{d}{dt}(t^3)$$

$$\frac{dz}{dt} = 4txy + 3t^2x^2$$

$$\frac{dz}{dt} = 4t^6 + 3t^6 = 7t^6$$

Example-12: Suppose that,

$$w = \sqrt{x^2 + y^2 + z^2}, x = \cos\theta, y = \sin\theta, z = \tan\theta$$

Use the chain rule to find $\frac{dw}{d\theta}$.

Solution:

$$\frac{dw}{d\theta} = \frac{\partial w}{\partial x} \frac{dx}{d\theta} + \frac{\partial w}{\partial y} \frac{dy}{d\theta} + \frac{\partial w}{\partial z} \frac{dz}{d\theta}$$

$$\begin{aligned} \frac{dw}{d\theta} &= \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2x)(-\sin\theta) + \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2y)(\cos\theta) \\ &\quad + \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2z)(\sec^2\theta) \end{aligned}$$

Chain Rule for Partial Derivative: If $x = x(u, v)$ and $y = y(u, v)$ have first-order partial derivatives at the point (u, v) , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(u, v), y(u, v))$, then $z = z(u, v)$ has first order partial derivatives at the point (u, v) given by

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \text{ and } \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

Example-13: $z = e^{xy}, x = 2u + v, y = u/v$

Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ by using chain rule.

$$\begin{aligned} \text{Solution: } \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \\ &= \frac{\partial}{\partial x}(e^{xy}) \frac{\partial}{\partial u}(2u + v) + \frac{\partial}{\partial y}(e^{xy}) \frac{\partial}{\partial u}(u/v) \\ &= 2ye^{xy} + xe^{xy} \frac{1}{v} \\ &= (2y + \frac{x}{v})e^{xy} \\ &= (2u/v + \frac{2u + v}{v})e^{(u/v)(2u+v)} \\ &= (\frac{4u}{v} + 1)e^{(u/v)(2u+v)} \end{aligned}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$\begin{aligned}
&= \frac{\partial}{\partial x}(e^{xy}) \frac{\partial}{\partial v}(2u + v) + \frac{\partial}{\partial y}(e^{xy}) \frac{\partial}{\partial v}(u/v) \\
&= ye^{xy} + xe^{xy}(-\frac{u}{v^2}) \\
&= (y - \frac{xu}{v^2})e^{xy} \\
&= [\frac{u}{v} - (2u + v)\frac{u}{v^2}]e^{(u/v)(2u+v)} \\
&= -\frac{u^2}{v^2}e^{(u/v)(2u+v)}
\end{aligned}$$

Example-14: $w = e^{xyz}$, $x = 3u + v$, $y = 3u - v$, $z = u^2v$

Find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ by using chain rule.

Solution: $\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial u}$

$$\begin{aligned}
&= \frac{\partial}{\partial x}(e^{xyz}) \frac{\partial}{\partial u}(3u + v) + \frac{\partial}{\partial y}(e^{xyz}) \frac{\partial}{\partial u}(3u - v) + \frac{\partial}{\partial z}(e^{xyz}) \frac{\partial}{\partial u}(u^2v) \\
&= 3ye^{xyz} + 3xe^{xyz} + 2uvxye^{xyz} \\
&= (3yz + 3xz + 2uvxy)e^{xyz} \\
\frac{\partial w}{\partial v} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial v} \\
&= \frac{\partial}{\partial x}(e^{xyz}) \frac{\partial}{\partial v}(3u + v) + \frac{\partial}{\partial y}(e^{xyz}) \frac{\partial}{\partial v}(3u - v) + \frac{\partial}{\partial z}(e^{xyz}) \frac{\partial}{\partial v}(u^2v) \\
&= ye^{xyz} - xe^{xyz} + u^2xye^{xyz} \\
&= (yz - xz + u^2xy)e^{xyz}
\end{aligned}$$

Directional Derivative: If $f(x, y)$ is a function of x and y and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector then the directional derivative of $\mathbf{u}$ at (x_0, y_0) is denoted by $D_{\mathbf{u}}f(x_0, y_0)$ and defined by

$$D_{\mathbf{u}}f(x_0, y_0) = \frac{d}{ds} [f(x_0 + su_1, y_0 + su_2)]_{s=0}$$

Provided the derivatives exist.

Geometrically $D_u f(x_0, y_0)$ can be interpreted as the slope of the surface $z = f(x, y)$ in the direction of $\mathbf{u}$ at the point $(x_0, y_0, f(x_0, y_0))$. Usually the value of $D_u f(x_0, y_0)$ will depend on both the point (x_0, y_0) and the direction $\mathbf{u}$. Thus, at a fixed point the slope of the surface may vary with the direction. Analytically, the directional derivative represents the instantaneous rate of change of $f(x, y)$ with respect to distance in the direction of $\mathbf{u}$ at the point (x_0, y_0) .

Formula: If $f(x, y)$ is a function of x and y and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ then the directional derivative,

$$D_u f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2$$

Example-15: Find the directional derivative of $f(x, y) = e^{xy}$, at $(-2, 0)$ in the direction of unit vector that makes an angle $\frac{\pi}{3}$ with the positive x-axis also find the unit vector.

Solution: The partial derivative of $f(x, y)$

$$f_x(x, y) = ye^{xy} \quad \text{and} \quad f_y(x, y) = xe^{xy}$$

$$f_x(-2, 0) = 0 \quad \text{and} \quad f_y(-2, 0) = -2$$

The unit vector unit vector $\mathbf{u}$ that makes an angle $\frac{\pi}{3}$ with the positive x-axis is,

$$\mathbf{u} = \cos \frac{\pi}{3} \mathbf{i} + \sin \frac{\pi}{3} \mathbf{j} = \frac{1}{2} \mathbf{i} + \frac{\sqrt{3}}{2} \mathbf{j}$$

$$D_u f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2$$

$$= f_x(-2, 0) \cos \frac{\pi}{3} + f_y(-2, 0) \sin \frac{\pi}{3} = -\sqrt{3}$$

Example-16: Find the directional derivative of $f(x, y, z) = x^2y - z^3y + z$ at the point $(-1, 2, 0)$ in the direction of the vector $\mathbf{u} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

Solution: $f(x, y, z) = x^2y - z^3y + z$

The partial derivative of $f(x, y, z)$

$$f_x(x, y, z) = 2xy \quad \text{and} \quad f_y(x, y, z) = x^2 - z^3$$

$$f_x(-1, 2, 0) = -4 \quad \text{and} \quad f_y(-1, 2, 0) = 1$$

$$f_z(x, y, z) = -3z^2y + 1$$

$$f_z(-1, 2, 0) = 1$$

$$u = \frac{a}{|a|} = \frac{2}{3}i + \frac{1}{3}j - \frac{2}{3}k$$

Therefore the directional derivative

$$D_u f(x_0, y_0, z_0) = -4\left(\frac{2}{3}\right) + \frac{1}{3} - \frac{2}{3} = -3$$

Exercise: 1. If $f(x, y) = \sqrt{y+1} + \ln(x^2 - y)$. Find and sketch the natural domain.

2. Let $f(x, y, z) = \sqrt{2 - x^2 - y^2 - z^2}$. Find $f\left(0, \frac{1}{3}, -\frac{1}{3}\right)$ and the natural domain of f .

3. If $w = x^2 + y^2 - z^2$, and $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta - 2z \cos \varphi$

Find $\frac{\partial w}{\partial \rho}$ and $\frac{\partial w}{\partial \theta}$ by using chain rule.

4. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{2x^6 + y^2}$

5. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{3}{x^2 + 2y^2}$

6. Evaluate $\lim_{(x,y) \rightarrow (1,0)} \frac{(x-1)^2 \ln x}{(x-1)^2 + y^2}$

7. Find the directional derivative of $f(x, y, z) = xy + z^2$ at the point $(-3, 0, 4)$ in the direction of the vector $\mathbf{u} = i + j + k$

8. Find the directional derivative of $f(x, y, z) = e^{x+y+3z}$ at the point $(-2, 2, -1)$ in the direction of the vector $\mathbf{u} = 20i - 4j + 5k$

9. Find the directional derivative of $f(x, y) = e^{x+3y}$, at $(-2, 0)$ in the direction of unit vector that makes an angle $\frac{\pi}{4}$ with the positive x-axis also find the unit vector.

